

Revisiting common information

August 2024

1 paper contents

1. rate-distortion with multiple utility functions at once (can be deduced from usual rate-distortion using convexity)
2. rate-distortion with the wyner diagram
3. generalizing the above to some more diagrams (trees?)
4. discussing the simplest open problems: wyner with a utility function that sees all three outputs at once and then open problem from the literature
5. the product codebook idea for going from independent variables to coding. can maybe also say something a bit more general here. this gives a single-channel lower bound on the utility attainable with coding

see exercise 14.9 in network information theory for wyner rate-distortion

unfortunately it looks like the bound with independent variables could never have a converse, because there are setups where one can't make independent variables at the optimal rates